

MAYA'S CASE FILE

10:47 PM. Tomorrow is Maya's big science test. Your team has been shrunk to the size of a cell and sent into her brain to help her study, but her brain just went into LOCKDOWN. Every room on the screen is sealed. The codes to open them are hidden in this Case File.

How to play

1. Start the game on the laptop and choose **With the printed Case File**.
2. Click a room on the brain map. Its sealed door tells you which **page** of this Case File to solve.
3. Read the room's **field notes** on the screen. Use them to solve the page with a pencil.
4. Type the code into the door. Then beat the puzzle on screen to win **one letter**.
5. Collect all 8 letters, then unscramble them in the **frontal lobe** to escape.

Start with the **Brainstem**. Nothing else opens until life support is back online. After that, split up: some teammates work on paper while others play on screen.

Team name

Team members

Letters we've collected

Write each letter the screen gives you.

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Unscrambled code:

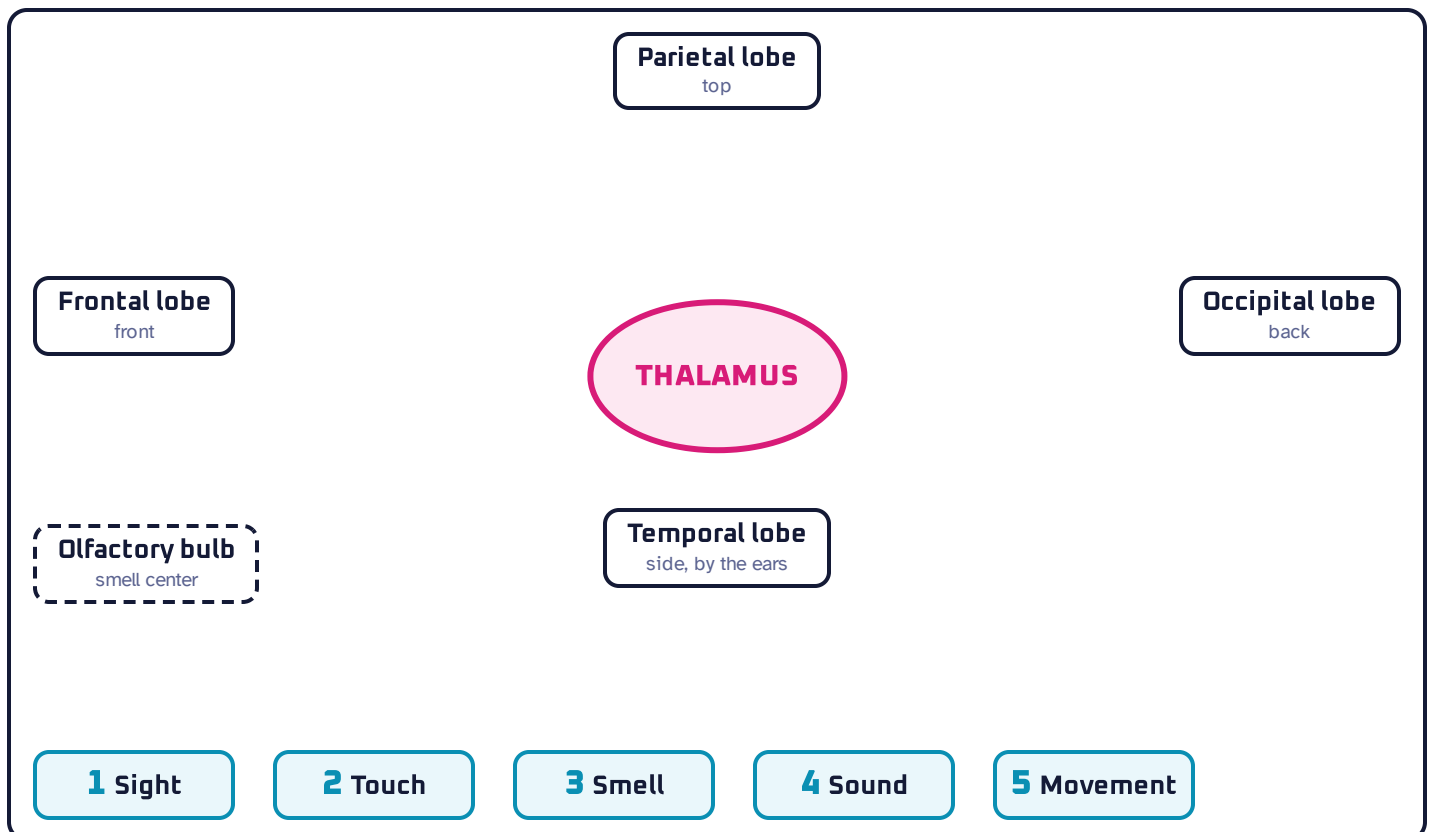
The brainstem runs some jobs **automatically**, without you deciding to do them. **Circle** the letter next to each automatic job. Read the circled letters from top to bottom.

1. Breathing while you sleep	(P)	5. Swallowing	(N)
2. Solving a math problem	(X)	6. Reading a book	(R)
3. Your heart beating	(O)	7. Sneezing	(S)
4. Choosing a snack	(T)	8. Texting a friend	(E)

DOOR CODE

Hint: it's a real part of the brainstem!

Draw a line from each sense **into the thalamus**, then **out to the lobe** that handles it. Write the direction each line leaves the thalamus (→ ↑ ↓ ←). A sense that never goes through the thalamus gets **no arrow**.



DOOR CODE

Sense 1

 2 3 4 5

The hypothalamus keeps your body balanced. **Draw a line** from each alert to the response that fixes it. One response isn't used. Write the letters in alert order.

1. Blood glucose dropping. Last meal was 6 hours ago. ☐

2. Body temp 35.9 °C. Waiting for the bus in the snow. ☐

3. Body temp 38.4 °C after running in gym class. ☐

4. Water level low. Blood is getting too salty. ☐

5. 10:30 PM. Dark room. The body clock says night. ☐

☐ Sweat

C

☐ Stress hormones

T

☐ Thirst

U

☐ Hunger

F

☐ Sleepiness

S

☐ Shiver

O

DOOR CODE

Fill in one letter per box. Every answer is in the amygdala's field notes. The **pink boxes** spell the door code, top to bottom.

1. A stress hormone the body releases

2. Its name comes from the Greek word for "almond"

3. This kind of breathing quiets the alarm

4. Paying attention to the present moment on purpose

DOOR CODE

THE VAULT • ONLY THE MEMORY KEEPER MAY LOOK

 3 sleep	 5 phone	 8 practice quiz	 2 soda	 9 video game	 1 water
 4 alarm clock	 7 candy	 0 headphones	 6 calm breathing	 5 TV	 2 pizza

× — — FOLD THE TOP HALF BACK ALONG THIS LINE BEFORE YOU START — —

Vault rules. Fold the top half of this page back so nobody can see it. Pick a **Memory Keeper**. The keeper unfolds the vault, looks for **20 seconds**, then folds it closed again. No writing while it's open! The keeper then tells the team what they remember. Want another look? Switch keepers, but first the new keeper has to say one brain fact out loud.

Which numbers were next to the four memory boosters?

The boosters are listed in the hippocampus field notes on screen. Keep them in that order.

1. Sleep

2. Practice

3. Water

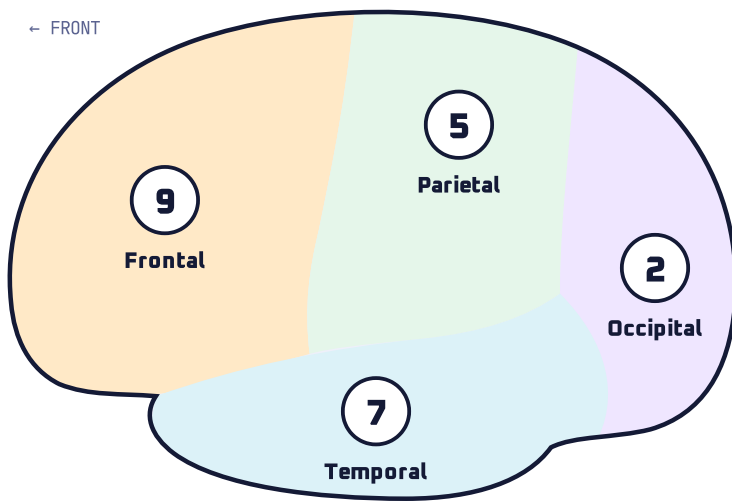
4. Calm

Memory tricks that work

Chunking: group numbers into pairs. **Rehearsal:** repeat them out loud. **Teamwork:** each person remembers one piece.

Linking: connect a number to a picture ("the moon has 3 craters").

Each lobe on this map has a number. Write the number of the lobe that does each job. The temporal lobes' field notes describe all four lobes.



1. Hearing a song on the radio

2. Seeing a friend wave at you

3. Feeling a pencil in your hand

4. Planning your homework

5. Understanding what your teacher says

DOOR CODE

Only **four** of these structures are on the limbic system team. **Circle** them (the field notes name them), then add up their numbers.

Brainstem

143

Amygdala

128

Cerebellum

91

Hypothalamus

74

Occipital lobe

212

Cingulate gyrus

180

Spinal cord

57

Hippocampus

205

DOOR CODE

_____ + _____ + _____ + _____ =

Maya's neurons send chemical messages. For each situation, **circle** the messenger that fits: ▲ **Dopamine** (reward, motivation, movement), ● **Serotonin** (mood, sleep, appetite), or ■ **GABA** (the brakes). The circled letters spell the door code.



1. Maya finally beats a level she's been stuck on, and can't wait to play again.

V

K

T

2. A good night's sleep and morning sunlight keep Maya's mood steady.

O

E

A

3. Maya's racing thoughts slow down so she can fall asleep.

R

N

S

4. This messenger is mostly made in the gut and helps control appetite.

U

I

M

5. Maya checks off her first homework task and feels motivated to do the next.

C

P

D

6. This messenger helps Maya's muscles move smoothly when she shoots a basketball.

L

B

H

7. The brakes: this messenger tells overexcited neurons to fire less.

Y

W

E

DOOR CODE

Hint: it's a word from the field notes.

How neurons talk

An electrical pulse runs down the neuron → tiny bubbles called vesicles release neurotransmitters → they cross the gap (the synapse) → they fit into receptors on the next neuron, like a key in a lock.